



Strategic Location Selection of Military Airports under Uncertainty: A Hybrid Multi-Criteria Decision-Making Approach

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Abstract: The strategic siting of a military airport constitutes a high-stakes planning problem characterized by complex trade-offs, long-term operational consequences, and pronounced uncertainty in expert judgment. In contrast to civilian airport planning, where economic efficiency and environmental externalities are typically prioritized, military airport location decisions are governed by additional requirements related to operational security, survivability, logistical resilience, and future capacity expansion. To address these challenges, a hybrid Multi-Criteria Decision-Making (MCDM) framework is proposed for the systematic evaluation and selection of military airport locations under uncertainty. Six core criteria and their associated sub-criteria, reflecting operational, strategic, technical, and infrastructural considerations, were identified through expert consultation and domain analysis. Criteria weights were derived using the Defining Interrelationships Between Ranked Criteria II (DIBR II) method and its Fuzzy, Grey, and Rough extensions, enabling the explicit modelling of vagueness, incompleteness, and ambiguity inherent in subjective assessments. Expert evaluations were aggregated using the Einstein Weighted Arithmetic Average (EWAA) operator, which accommodates heterogeneous levels of expertise and mitigates dominance bias. Alternative locations were subsequently ranked using the Weighted Aggregated Sum Product Assessment (WASPAS) method, allowing for flexible integration of additive and multiplicative aggregation schemes. The robustness of the obtained rankings was examined through a sensitivity analysis of the WASPAS aggregation parameter λ , confirming that variations in the aggregation structure do not alter the identification of the optimal and least-preferred alternatives. Furthermore, a comparative analysis with five established MCDM techniques revealed a high degree of rank correlation, thereby reinforcing the internal consistency and reliability of the proposed framework. The results demonstrate that the integration of uncertainty theories with advanced MCDM techniques provides a rigorous and adaptable decision-support tool for military infrastructure planning. Owing to its modular structure and methodological generality, the proposed framework can be readily adapted to diverse geographical settings, operational doctrines, and security environments, offering practical value for strategic decision-making in the defense sector.

Keywords: Multi-Criteria Decision-Making; Military; Airport; Location; Fuzzy; Defining Interrelationships Between Ranked Criteria II; Einstein Weighted Arithmetic Average; Weighted Aggregated Sum Product Assessment

1 Introduction

The selection of a military airport location is one of the most important strategic infrastructure decisions, as it directly affects national security, operational readiness, and the long-term development of defense capabilities. Such projects require consideration of a wide range of criteria, from natural conditions, through connectivity to transportation networks, to social and environmental impacts. Unlike civilian airports, where accessibility, commercial viability, and environmental acceptability are dominant criteria, military airports require a specific set of factors that include security, discretion, resistance to attacks, logistical connectivity, and capacity expansion. Such projects must take into account not only technical and infrastructural aspects, but also the broader strategic context, including different circumstances and potential security threats [1–4].

A review of the literature shows that the criteria for choosing the location of a military airport are evolving from engineering and strategic bases to modern multi-criteria methods. Cozens [1] provides a historical and engineering framework for understanding the differences between civilian and military airports. For civilian airports, it emphasizes

accessibility, traffic connectivity, stable meteorological conditions, and minimal environmental impact, while for military airports, it emphasizes operational efficiency, discretion, resistance to attacks, and availability of resources. In addition to specific criteria, Cozens [1] identifies general factors such as soil quality, drainage and minimizing the removal of existing structures, which are common to all types of airports. This analysis shows the continuity of criteria over time and their evolution from an engineering approach to modern methods.

Sennaroglu and Celebi [2] focus on the site selection of a military airport through the integration of Analytic Hierarchy Process (AHP), Preference Ranking Organization Method for Enrichment Evaluations (PROMETHEE) and Visekriterijumska Optimizacija I Kompromisno Resenje (VIKOR) methods. This approach allows criteria to be first quantified and ranked using AHP, and then verified using outlier and trade-off methods, thus achieving robustness of decisions. The authors emphasize the importance of security risks, logistical connectivity, proximity to military units, and expansion opportunities, demonstrating that military airports require specific criteria that go beyond the commercial needs of civilian airports.

Erkan and Elsharida [3] systematize the methods used for airport location selection, including classical location models, Multi-Criteria Decision-Making methods (MCDM) such as AHP, Technique for Order Preference by Similarity to Ideal Solution (TOPSIS), PROMETHEE, VIKOR and their integration with geographic information systems (GIS) tools. Their research highlights basic indicators such as cost, accessibility, capacity, environmental factors, topography and safety, which can be applied in a military context with additional requirements. The authors conclude that the most effective approach is a combination of MCDM methods with spatial analysis, as this provides a realistic assessment of terrain suitability and infrastructure constraints.

Rangsaritvorakarn [4] presents a review of 16 relevant studies and systematizes the criteria used in airport location selection, with environmental factors such as air and noise pollution, as well as connectivity to major transport networks, being most frequently highlighted, while socio-economic and safety factors are present to a lesser extent, and distance from other airports is least represented; the contribution of the work is that it provides an overview of the frequency of criteria and highlights the dominance of environmental and infrastructure aspects.

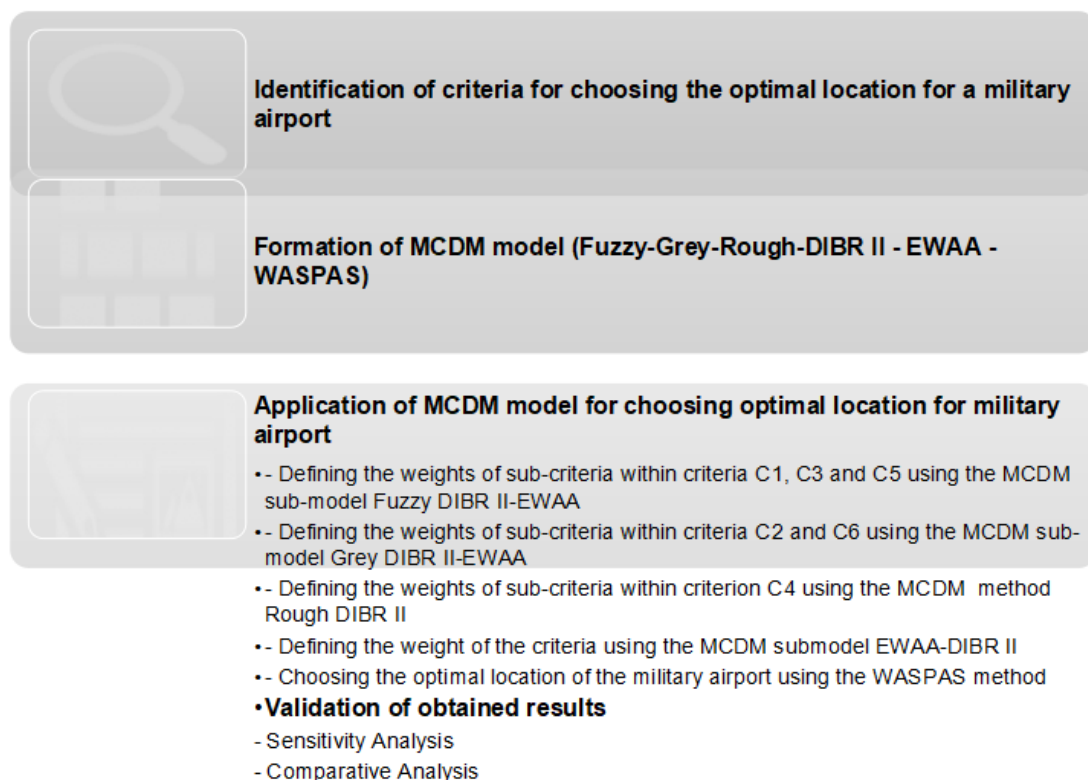


Figure 1. Research algorithm

Note: Multi-Criteria Decision-Making (MCDM); Defining Interrelationships Between Ranked Criteria II (DIBR II); Einstein Weighted Arithmetic Average (EWAA); Weighted Aggregated Sum Product Assessment (WASPAS)

Based on these findings, it can be concluded that the choice of a military airport requires an MCDM approach that combines technical, security, logistical and social criteria. This approach has become a key tool for solving complex military decision-making problems, as it enables systematic consideration of various factors and provides a more reliable framework for decision-making through hybrid approaches and integration with modern technologies [5]. Also,

the application of MCDM methods, through the simultaneous analysis of several mutually conflicting criteria, enables systematic and objective decision-making in various areas such as digital business and e-commerce strategies [6], consumer behavior and marketing of green products [7], management of natural disasters [8], renewable energy [9], human capital development [10], medicine-diagnosis, treatment, and prevention of disease [11], the insurance industry [12], planning of educational infrastructure [13], ecology [14], economy [15], etc.

There is no one universal theory that can cover all aspects of such a complex problem, so in practice it is shown that a hybrid approach is the most effective. This is precisely why contemporary research [16–18] is increasingly moving towards combining theories within the MCDM model. The nature of the problem requires the application of different theoretical frameworks: Fuzzy theory is suitable for subjective and linguistic assessments, that is, it enables the modeling of imprecise boundaries and continuous grades [19, 20], grey theory for interval and sparse data [21, 22], while rough theory enables the treatment of incomplete knowledge [23, 24]. The combination of fuzzy, grey and rough theories within a multi-criteria model provides a transparent and robust framework for decision-making on the optimal location of a military airport.

In order to carry out this research, the algorithm presented in Figure 1 was formed.

First of all, it is necessary, based on existing research, to identify the criteria that condition the subject choice. After that, the weights of both sub-criteria and criteria should be defined. After obtaining the weight coefficients of the criteria, the selection of the optimal alternative is approached. In order to validate the results obtained, a sensitivity and comparative analysis will be performed, as well as to discuss the results of the subject analyses.

2 Methodology

For the purposes of this research, the Defining Interrelationships Between Ranked criteria II (DIBR II) method [25] was used to determine the weight coefficients of the criteria, as well as its improved version using fuzzy [26], grey [27] and rough theories [28]. The Einstein Weighted Arithmetic Average (EWAA) operator [29, 30] was used to aggregate expert opinions, taking into account their competence coefficients, while the Weighted Aggregates Sum Product Assessment (WASPAS) method [31] was used to select the optimal location of the military airport. In the following, the methods used and the operator are described.

2.1 Defining Interrelationships Between Ranked Criteria II Method

This method, published for the first time in 2023 [25], has so far found its application in various areas, such as sustainable energy and energy systems [32], military and defense [33, 34], lean management and organizational systems [35], risk management and protection from natural disasters [36], etc. DIBR II is a method based on pairwise comparisons of adjacent criteria, where their relative importance is used to establish a complete network of interrelationships across a range of ranked criteria. Based on these relationships, the consistent weights of all criteria are directly calculated, with a final check of the logic of the relationship with an acceptable deviation [25]. Given that the mathematical formulation of the DIBR II method is presented in numerous publications, the pseudocode of the method according to Božanić and Pamucar [25] is presented in Figure 2.

In this research, for the purposes of determining the weights of the criteria (ω^C), the DIBR II method was used, while for defining the weights of the sub-criteria, improved versions of this method were used using the fuzzy, grey and rough theories. For sub-criteria that require subjective and linguistic assessments, that is, modeling of imprecise boundaries, the Fuzzy DIBR II method was used [37]. Triangular fuzzy numbers when defining the relationship between sub-criteria in this research were created in the same way as by Tešić et al. [37], using the level of experts' confidence. Defuzzification of fuzzy numbers is performed using Eq. (1):

$$\text{def}(\tilde{H}) = ((h_3 - h_1) + (h_2 - h_1)) / 3 + h_1 \quad (1)$$

where, $\tilde{H}$ is a triangular fuzzy number that has the following appearance $\tilde{H} = (h_1, h_2, h_3)$ and the elements of the number represent the left, middle, and right distributions. The middle distribution represents the point where the membership function has the value 1.

For subcriteria representing interval and sparse data, the Grey DIBR II method was used [38]. Interval gray numbers when defining the relationship between sub-criteria in this research were created in the same way as by Tešić et al. [38], using the level of experts' confidence. Obtaining crisp values from grey numbers is done by applying Eq. (2):

$$\text{crisp}(\otimes H) = (1 - \psi) \cdot \underline{h} + \psi \cdot \bar{h} \quad (2)$$

where, $\otimes H$ is an interval grey number that can be represented as $\otimes H = (\underline{h}, \bar{h})$, consisting of a lower limit and an upper limit; ψ is the whitening coefficient, with $\psi \in \{0, 1\}$ [38].

For the purposes of determining the relationship between sub-criteria where treatment of incomplete knowledge is necessary, the Rough DIBR II method [39] was used. Rough numbers $RN(H) = [\underline{h}, \bar{h}]$ in this research were created

in the same way as by Tešić and Khalilzadeh [39], so that the lowest comparison value given by the experts is adopted for the lower limit ($\underline{h}$), while the highest value is adopted for the upper limit ($\bar{h}$). Obtaining crisp values from rough numbers is done by applying Eqs. (3)–(5) [39, 40]:

$$RN(H) = [\underline{h}, \bar{h}] \Rightarrow \left\{ \begin{array}{l} \underline{h} = \frac{\underline{h} - \min_i \{h\}}{\max_i \{h\} - \min_i \{h\}} \\ \bar{h} = \frac{\bar{h} - \min_i \{h\}}{\max_i \{h\} - \min_i \{h\}} \end{array} \right\} \quad (3)$$

$$H^N = \frac{\underline{h} \bullet \{1 - \underline{h}\} + \bar{h} \bullet \bar{h}}{1 - \underline{h} + \bar{h}} \quad (4)$$

$$H^{\text{crisp}} = \min_i \{h\} + H^N \bullet [\max_i \{\bar{h}\} - \min_i \{h\}] \quad (5)$$

```
function DIBR_II(criteria):
    // Step 1: Define criteria
    K = criteria

    // Step 2: Rank criteria
    ranked = sort_by_importance(K)

    // Step 3: Define relationships between adjacent criteria
    relations = {}
    for i in range(0, len(ranked) - 1):
        relations[(ranked[i], ranked[i+1])] = define_importance_ratio(ranked[i], ranked[i+1])

    // Also define relationship between first and last criterion
    relations[(ranked[0], ranked[-1])] = define_importance_ratio(ranked[0], ranked[-1])

    // Step 4: Link most important criterion to all others
    links = {}
    for criterion in ranked:
        links[criterion] = derive_relationship(ranked[0], criterion, relations)

    // Step 5: Assign weight to the most important criterion
    w_main = compute_main_weight(links)

    // Step 6: Compute weights of remaining criteria
    weights = {}
    for criterion in ranked:
        weights[criterion] = compute_weight(criterion, w_main, links)

    // Step 7: Consistency check
    for criterion in ranked:
        deviation = compute_deviation(criterion, weights, links)
        if deviation > 0.10:
            raise Error("Inconsistent relationships detected. Redefine adjacent relations.")

    return weights
```

Figure 2. Pseudocode of the Defining Interrelationships Between Ranked Criteria II (DIBR II) method

2.2 Einstein Weighted Arithmetic Average Operator

The EWAA operator is an advanced aggregation operator based on Einstein's operations, which introduce a non-linear transformation into the aggregation process and thus enable a more robust aggregation of expert opinions [29, 30]. In this research, it is used to aggregate expert opinions, i.e., comparisons of adjacent criteria by experts when obtaining criterion weight coefficients. The mathematical formulation of this operator is given by Eq. (6) [29, 30].

$$EWAA \{b_1, b_2, \dots, b_j\} = \sum_{j=1}^E b_j^e \frac{\prod_{j=1}^E (1 + f(b_j^E))^{\omega^E} - \prod_{j=1}^E (1 - f(b_j^E))^{\omega^E}}{\prod_{j=1}^E (1 + f(b_j^E))^{\omega^E} + \prod_{j=1}^E (1 - f(b_j^E))^{\omega^E}} \quad (6)$$

where, ω^E represents the weight coefficient of the expert's competence.

2.3 Weighted Aggregated Sum Product Assessment Method

WASPAS is an MCDM method that combines the advantages of the Weighted Sum Model (WSM) and the Weighted Product Model (WPM) to increase the accuracy of alternative ranking. It calculates the linear (WSM) and multiplicative (WPM) scores and then combines them into a single aggregation function that provides more stable and reliable results [41]. This method, since its publication in 2012 [31], has been applied, both in its basic (crisp) and modified versions, in various decision-making problems in different areas, such as industrial engineering and supply chain management [42], finance and investment management [43], energy and renewable energy sources with sustainable agriculture and land management [44], software engineering [45], military and defense [46], healthcare management [47], etc. In addition to the above [44, 46], this method was also used in numerous other papers to select different locations, for example [48, 49]. Given that the mathematical formulation of the WASPAS method is presented in numerous publications, the pseudocode of the method according to the study [31] is presented in Figure 3.

```
function DIBR_II(criteria):

    // Step 1: Define criteria
    K = criteria

    // Step 2: Rank criteria
    ranked = sort_by_importance(K)

    // Step 3: Define relationships between adjacent criteria
    relations = {}
    for i in range(0, len(ranked) - 1):
        relations[(ranked[i], ranked[i+1])] = define_importance_ratio(ranked[i], ranked[i+1])

    // Also define relationship between first and last criterion
    relations[(ranked[0], ranked[-1])] = define_importance_ratio(ranked[0], ranked[-1])

    // Step 4: Link most important criterion to all others
    links = {}
    for criterion in ranked:
        links[criterion] = derive_relationship(ranked[0], criterion, relations)

    // Step 5: Assign weight to the most important criterion
    w_main = compute_main_weight(links)

    // Step 6: Compute weights of remaining criteria
    weights = {}
    for criterion in ranked:
        weights[criterion] = compute_weight(criterion, w_main, links)

    // Step 7: Consistency check
    for criterion in ranked:
        deviation = compute_deviation(criterion, weights, links)
        if deviation > 0.10:
            raise Error("Inconsistent relationships detected. Redefine adjacent relations.")

    return weights
```

Figure 3. Pseudocode of the Weighted Aggregated Sum Product Assessment (WASPAS) method

3 Choosing the Optimal Location for a Military Airport

By analyzing previous research [1–4], the basic criteria can be grouped into six categories with clearly defined sub-criteria (Table 1), which provides a clear and mathematically structured framework for the application of MCDM methods in real military scenarios, that is, choosing the optimal location for a military airport. The sub-criteria are classified into Benefiting-type (B) and Cost-type (C) criteria.

A total of 12 experts $E = \{E_1, E_2, \dots, E_{12}\}$ from the field of research were hired for the purposes of this research, with the following coefficients of expert competencies ω^E (Table 2).

The experts agreed that the sub-criteria and the criteria in Table 1 are arranged by importance. Given that the sub-criteria within criteria C1, C3 and C5 include subjective assessments and linguistic expressions, that is, they require modeling of imprecise boundaries and continuous grades for the calculation of their weights, the Fuzzy DIBR II method was used. Following the algorithm of the above method, the experts assessed the relationship between the sub-criteria within each of the three above criteria (Table 3).

By applying the mathematical apparatus of the Fuzzy DIBR II method, for each of the experts, the weights of each of the sub-criteria within the criteria C1, C3 and C5 were obtained (Table 4). Fuzzy values were previously converted

to crisp using Eq. (1).

To obtain the unique criteria weights that will later be used in the computation of the optimal alternative, the weights of the criteria obtained for each of the experts (Table 4) were aggregated using the EWAA operator, where the weight coefficients of the experts' competencies were taken into account (Table 5), and the following values of the weight of each of the sub-criteria within criteria C1, C3 and C5 were obtained.

In the following, the weights of other sub-criteria within other criteria were calculated. Sub-criteria within criteria C2 and C6 imply partial, scarce or unclear numerical data, especially in the early stages of planning, so the calculation of their weights is performed using the Grey DIBR II method. In order to obtain the weighting coefficients of the sub-criteria within the subject criteria, the experts evaluated the relationship between them in accordance with the methodology of the Grey DIBR II method (Table 6).

Using the Grey DIBR II method, the weights of the sub-criteria were obtained for each of the experts, presented in Table 7. Grey values were previously converted to crisp using Eq. (2) and $\psi = 0.5$.

By applying the EWAA operator, following the same principle as in the previous case, the final weight coefficients of the sub-criteria for criteria C2 and C6 were obtained (Table 8).

Table 1. Identified criteria for choosing a location for a military airport

Criterion/Sub-Criterion	Description	Type
C1-Security criteria		
C11-Security risks	Assessment of terrorist threats, security standards	C
C12-Resistance to attacks	Multiple runways for aircraft dispersion, minimization of bomb damage	B
C13-Location discretion	Distance from populated areas to reduce risk and maintain secrecy	B
C2-Operational efficiency and military logistics		
C21-Proximity to military units and bases	To ensure the rapid use of the airport by military units	C
C22-Coordination with other military airports	Network support and interoperability	B
C23-Access to key points	Highways, railways, ports for rapid mobilization	B
C24-Accessibility to resources	Water, energy, building materials, communication infrastructure	B
C3-Capacity and expandability		
C31-Expansion space possibilities	Land that enables future capacity expansion	B
C32-Capacity to increase operations	The possibility of increasing the number of flights, storing equipment and fuel	B
C33-Infrastructure flexibility	Adaptation of new runways and facilities in accordance with military needs	B
C4-Geographical and climatic conditions		
C41-Topography and geological structure	The suitability of the terrain for military operations, the stability of the ground	B
C42-Climatic factors	Fog, frost, storms, floods – impact on operability	C
C43-Natural obstacles	Avoiding hills, tall buildings or terrain that makes access difficult	C
C5-Environmental and social factors		
C51-Impact on the local community	Noise and pollution	C
C52-Risks for the population	Health risks due to emissions and noise	C
C53-Balance with civilian needs	Possibility of dual use (military-civilian airport) but also potential economic benefits	B
C6-Costs and sustainability		
C61-Construction and maintenance costs	Financial costs of the project	C
C62-Efficient solution of drainage and water	Minimization of infrastructure problems	B
C63-Minimal need to remove existing facilities	Reduction of additional costs due to the removal of existing facilities	C

Table 2. Coefficients of expert competencies

Expert	E1	E2	E3	E4	E5	E6	E7	E8	E9	E10	E11	E12
ω^E	0.0870	0.0820	0.0840	0.0860	0.0830	0.0810	0.0800	0.0850	0.0825	0.0835	0.0845	0.0815

Table 3. Relationships between sub-criteria within criteria C1, C3 and C5

Sub-Criterion Relationship/ Experts	C11-C12	C12-C13	C11-C13	C31-C32	C32-C33	C31-C33	C51-C52	C52-C53	C51-C53	Degree of Confidence
E1	1.2	1.3	1.5	1.7	1	1.7	1.6	1	1.6	0.9
E2	1.3	1	1.3	1.7	1	1.7	1.5	1	1.5	1
E3	1.3	1.2	1.5	1.75	1	1.75	1.5	1	1.5	1
E4	1.2	1.1	1.3	1.7	1	1.7	1.4	1.1	1.5	0.9
E5	1.3	1.1	1.4	1.6	1.1	1.7	1.6	1	1.6	0.9
E6	1.2	1.2	1.4	1.75	1	1.75	1.5	1	1.5	1
E7	1.1	1.3	1.4	1.75	1	1.75	1.5	1	1.5	1
E8	1.2	1.3	1.5	1.6	1.1	1.7	1.6	1.1	1.7	0.8
E9	1.2	1.3	1.5	1.7	1	1.7	1.5	1	1.5	1
E10	1.2	1.1	1.3	1.75	1.1	1.7	1.4	1.1	1.5	0.9
E11	1.2	1.3	1.5	1.8	1	1.8	1.4	1.1	1.5	0.9
E12	1.3	1.2	1.5	1.7	1	1.7	1.5	1	1.5	1

Table 4. Weights of sub-criteria within criteria C1, C3 and C5 obtained for each of the experts

Sub-Criteria/ Experts	C11	C12	C13	C31	C32	C33	C51	C52	C53
E1	0.403243	0.336314	0.260443	0.462183	0.272417	0.265401	0.447186	0.28002	0.272794
E2	0.393939	0.303030	0.303030	0.459459	0.27027	0.27027	0.428571	0.285714	0.285714
E3	0.414894	0.319149	0.265957	0.466667	0.266667	0.266667	0.428571	0.285714	0.285714
E4	0.385452	0.321410	0.293139	0.462183	0.272417	0.265401	0.422437	0.302023	0.27554
E5	0.404520	0.311414	0.284066	0.455207	0.284851	0.259942	0.447186	0.28002	0.272794
E6	0.395604	0.329670	0.274725	0.466667	0.266667	0.266667	0.428571	0.285714	0.285714
E7	0.383378	0.348525	0.268097	0.466667	0.266667	0.266667	0.428571	0.285714	0.285714
E8	0.403197	0.333478	0.263325	0.456078	0.286971	0.25695	0.456078	0.286971	0.25695
E9	0.404145	0.336788	0.259067	0.459459	0.27027	0.27027	0.428571	0.285714	0.285714
E10	0.385452	0.321410	0.293139	0.477436	0.273205	0.249359	0.422437	0.302023	0.27554
E11	0.403243	0.336314	0.260443	0.476386	0.265216	0.258398	0.422437	0.302023	0.27554
E12	0.414894	0.319149	0.265957	0.459459	0.27027	0.27027	0.428571	0.285714	0.285714

Table 5. Aggregated weights of sub-criteria within criteria C1, C3 and C5

Sub-Criteria	C11	C12	C13	C31	C32	C33	C51	C52	C53
ω^C	0.399363	0.326391	0.274256	0.463987	0.272207	0.263806	0.432515	0.288996	0.278489

Table 6. Relationships between sub-criteria within criteria C2 and C6

Sub-Criterion Relationship/ Experts	C21-C22	C22-C23	C23-C24	C21-C24	C61-C62	C62-C63	C61-C63	Degree of Confidence
E1	1.7	1.1	2.5	4	2.4	1.1	2.6	0.9
E2	1.6	1	2.5	4	2.3	1	2.3	1
E3	1.5	1	2.5	4	2.3	1	2.3	1
E4	1.6	1	2.6	4	2.2	1	2.2	0.9
E5	1.5	1.1	2.6	3.5	2.3	1.1	2.5	0.9
E6	1.6	1	2.5	4	2.3	1	2.3	1
E7	1.6	1	2.5	4	2.2	1	2.2	1
E8	1.5	1.2	2.6	3.5	2.5	1.2	2.5	0.8
E9	1.6	1	2.5	4	2.3	1	2.3	1
E10	1.5	1.1	2.4	3.5	2.3	1.1	2.5	0.9
E11	1.7	1	2.5	4	2.4	1	2.4	0.9
E12	1.6	1	2.5	4	2.3	1	2.3	1

Table 7. Weights of sub-criteria within criteria C2 and C6 obtained for each of the experts

Sub-Criteria/ Experts	C21	C22	C23	C24	C61	C62	C63
E1	0.386507	0.251225	0.251225	0.111043	0.520610	0.239695	0.239695
E2	0.400000	0.250000	0.250000	0.100000	0.534884	0.232558	0.232558
E3	0.384615	0.256410	0.256410	0.102564	0.534884	0.232558	0.232558
E4	0.373868	0.258198	0.258198	0.109736	0.498869	0.250566	0.250566
E5	0.358887	0.264376	0.264376	0.112361	0.509980	0.245010	0.245010
E6	0.400000	0.250000	0.250000	0.100000	0.534884	0.232558	0.232558
E7	0.400000	0.250000	0.250000	0.100000	0.523810	0.238095	0.238095
E8	0.332461	0.270327	0.270327	0.126886	0.506098	0.246951	0.246951
E9	0.400000	0.250000	0.250000	0.100000	0.534884	0.232558	0.232558
E10	0.355558	0.261923	0.261923	0.120595	0.509980	0.245010	0.245010
E11	0.386507	0.251225	0.251225	0.111043	0.520610	0.239695	0.239695
E12	0.400000	0.250000	0.250000	0.100000	0.534884	0.232558	0.232558

Table 8. Aggregated weights of sub-criteria within criteria C2 and C6

Sub-Criteria	C21	C22	C23	C24	C61	C62	C63
ω^C	0.381285	0.255367	0.255367	0.107981	0.521860	0.239070	0.239070

Table 9. Relationships between sub-criteria within criteria C4

Sub-Criterion Relationship/ Experts	C41-C42	C42-C43	C41-C43
E1	1.7	1.1	4
E2	1.6	1	4
E3	1.5	1	4
E4	1.6	1	4
E5	1.5	1.1	3.5
E6	1.6	1	4
E7	1.6	1	4
E8	1.5	1.2	3.5
E9	1.6	1	4
E10	1.5	1.1	3.5
E11	1.7	1	4
E12	1.6	1	4
Rough sub-criterion relationship	[1.2, 1.5]	[1.5, 1.8]	[1.8, 2.7]

Table 10. Final values of weights of sub-criteria within criteria C4

Sub-Criteria	C41	C42	C43
ω^C	0.471664	0.345258	0.183078

It is still necessary to determine the values of sub-criteria within criterion C4. Incomplete or missing data, where there is a border between certain and uncertain knowledge in sub-criteria within criterion C4, are treated using the rough theory, that is, the weights of the sub criteria in this criterion are determined using the Rough DIBR II method. Respecting the algorithm of the method, the experts evaluated the relationship between the sub-criteria, and the results of the comparison are presented in Table 9. Rough values of the ratio between the sub-criteria were obtained by taking the minimum value of the comparison by experts as the lower limit for each ratio, and the maximum value as the upper limit, in accordance with the algorithm of the Rough DIBR II method.

Using the mathematical apparatus of the Rough DIBR II method, the weights of the sub-criteria (within criterion C4) were obtained and shown in Table 10. Rough values were previously converted to crisp using Eqs. (3)–(5).

Given that the weight values of all sub-criteria within each of the criteria, i.e., their local values, are obtained, it is necessary to obtain the weight values of all criteria, in order to obtain the global weight values of all sub-criteria, with the aim of their successful implementation in the TRUST method and the selection of the optimal alternative. The above will be carried out using the DIBR II method. The experts primarily agreed that the criteria were ranked

according to importance as follows $C1 > C2 > C4 > C3 > C6 > C5$. The same experts evaluated the relationship between the criteria, and the comparison values are given in Table 11. Comparison values are aggregated values using the EWAA operator and are presented in the last row of this table.

Table 11. Relationships between criteria

Criterion Relationship/ Experts	C1-C2	C2-C4	C4-C3	C3-C6	C6-C5	C1-C5
E1	1.6	1.3	1	1.1	1.9	4
E2	1.5	1.3	1	1.2	1.8	5
E3	1.5	1.4	1.1	1.2	1.8	5
E4	1.4	1.5	1.1	1.1	1.9	4
E5	1.7	1.4	1	1	1.7	5
E6	1.6	1.3	1.1	1.1	1.9	5
E7	1.5	1.4	1.1	1.2	1.8	5
E8	1.4	1.5	1	1	1.9	5
E9	1.5	1.2	1.1	1.2	1.8	4
E10	1.4	1.3	1.1	1.2	1.9	5
E11	1.6	1.2	1.1	1.2	1.8	4
E12	1.5	1.2	1.1	1.2	1.9	5
Aggregated comparison values	1.516438	1.334106	1.066314	1.141035	1.842115	4.660328

Table 12. Final weights of criteria

Criteria	C1	C2	C3	C4	C5	C6
ω^C	0.308253	0.203275	0.142892	0.152368	0.067982	0.125230

Table 13. The global weight values of all sub-criteria

Criterion/Sub-Criterion	Criteria/Sub-Criteria Weight
C1-Security criteria	0.308253
C11-Security risks	0.123104
C12-Resistance to attacks	0.100610
C13-Location discretion	0.084539
C2-Operational efficiency and military logistics	0.203275
C21-Proximity to military units and bases	0.077505
C22-Coordination with other military airports	0.051910
C23-Access to key points	0.051910
C24-Accessibility to resources	0.021950
C3-Capacity and expandability	0.142892
C31-Expansion space possibilities	0.066300
C32-Capacity to increase operations	0.038896
C33-Infrastructure flexibility	0.037696
C4-Geographical and climatic conditions	0.152368
C41-Topography and geological structure	0.071867
C42-Climatic factors	0.052606
C43-Natural obstacles	0.027895
C5-Environmental and social factors	0.067982
C51-Impact on the local community	0.029403
C52-Risks for the population	0.019647
C53-Balance with civilian needs	0.018932
C6-Costs and sustainability	0.125230
C61-Construction and maintenance costs	0.065352
C62-Efficient solution of drainage and water	0.029939
C63-Minimal need to remove existing facilities	0.029939

By applying the DIBR II method to the aggregated comparison values from Table 11, the final values of the criterion weights are obtained (Table 12).

After the weights of the criteria were obtained, the global weight values of all sub-criteria were calculated (Table 13).

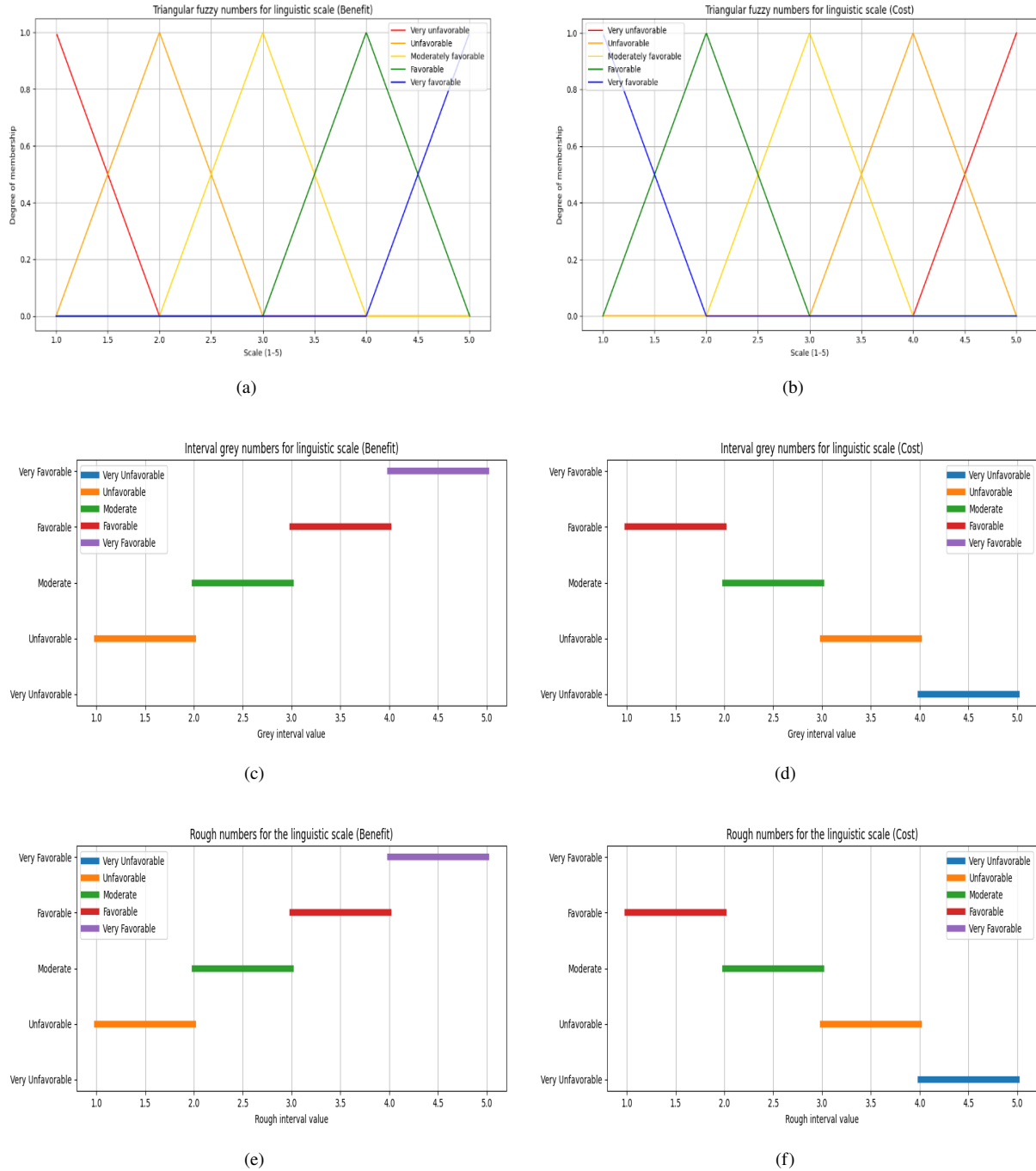


Figure 4. Linguistic scale for evaluating alternatives: (a) for criteria C12, C53 and C13; (b) for criteria C11, C51 and C52; (c) for criteria C22, C23, C24 and C62; (d) for criteria C21, C61 and C63; (e) for criterion C41; (f) for criteria C42 and C43

In order to choose the optimal alternative, and given that the criteria and sub-criteria are of a qualitative type, a linguistic scale was defined for evaluating the alternatives (Figure 4). Skala se sastoji iz sledecih lingvističkih deskriptora: Very favorable (VFA), Favorable (FAV), Moderately favorable (MFA), Unfavorable (UNF), Very unfavorable (VUN).

Using the linguistic scale (Figure 2), and in accordance with the type of criteria, the decision maker evaluated the five defined alternatives $A = (A_1, A_2, \dots, A_5)$ presented in the initial decision matrix (Table 14).

By converting linguistic descriptors into crisp values, applying Eqs. (1)–(5) the following numerical initial decision matrix is obtained (Table 15).

Using the mathematical apparatus of the WASPAS method, and a value of 0.5 of the parameter λ , the values of total relative importance and ranking of the alternatives presented in Figure 5 were obtained.

Table 14. Linguistic initial decision matrix

Criteria/Alternative	C11	C12	C13	C21	C22	C23	C24	C31	C32	C33
A1	FAV	MFA	VFA	MFA	MFA	FAV	MFA	UNF	MFA	VFA
A2	MFA	MFA	FAV	MFA	MFA	FAV	VFA	FAV	FAV	MFA
A3	MFA	VFA	MFA	VFA	FAV	VFA	FAV	VFA	MFA	VFA
A4	FAV	VFA	FAV	UNF	MFA	FAV	MFA	FAV	MFA	UNF
A5	FAV	FAV	MFA	MFA	FAV	MFA	FAV	MFA	VFA	MFA
Criteria/Alternative	C41	C42	C43	C51	C52	C53	C61	C62	C63	
A1	MFA	VFA	FAV	MFA	VFA	VFA	VFA	VFA	VFA	
A2	FAV	FAV	VFA	UNF	FAV	FAV	MFA	MFA	FAV	
A3	MFA	FAV	FAV	MFA	MFA	FAV	FAV	MFA	UNF	
A4	VFA	VFA	FAV	FAV	MFA	VFA	FAV	VFA	VFA	
A5	VFA	FAV	FAV	VFA	VFA	MFA	FAV	MFA	FAV	

Note: Very favorable (VFA); Favorable (FAV); Moderately favorable (MFA); Unfavorable (UNF); Very unfavorable (VUN).

Table 15. Numerical initial decision matrix

ω^C	0.123104	0.100610	0.084539	0.077505	0.051910	0.051910	0.021950	0.066300	0.038896	0.037696
Criteria Type	C	B	B	C	B	B	B	B	B	B
Criteria/Alternative	C11	C12	C13	C21	C22	C23	C24	C31	C32	C33
A1	2.00	3.00	4.67	2.80	2.80	3.80	2.80	2.00	3.00	4.67
A2	3.00	3.00	4.00	2.80	2.80	3.80	4.80	4.00	4.00	3.00
A3	3.00	4.67	3.00	1.00	3.80	4.80	3.80	4.67	3.00	4.67
A4	2.00	4.67	4.00	3.80	2.80	3.80	2.80	4.00	3.00	2.00
A5	2.00	4.00	3.00	2.80	3.80	2.80	3.80	3.00	4.67	3.00
ω^C	0.071867	0.052606	0.027895	0.029403	0.019647	0.018932	0.065352	0.029939	0.029939	
Criteria Type	B	C	C	C	C	B	C	B	C	
Criteria/Alternative	C41	C42	C43	C51	C52	C53	C61	C62	C63	
A1	2.80	1.00	1.90	3.00	1.00	4.67	1.00	4.80	1.00	
A2	3.70	1.90	1.00	4.00	2.00	4.00	2.80	2.80	1.80	
A3	2.80	1.90	1.90	3.00	3.00	4.00	1.80	2.80	3.80	
A4	4.60	1.00	1.90	2.00	3.00	4.67	1.80	4.80	1.00	
A5	4.60	1.90	1.90	1.00	1.00	3.00	1.80	2.80	1.80	

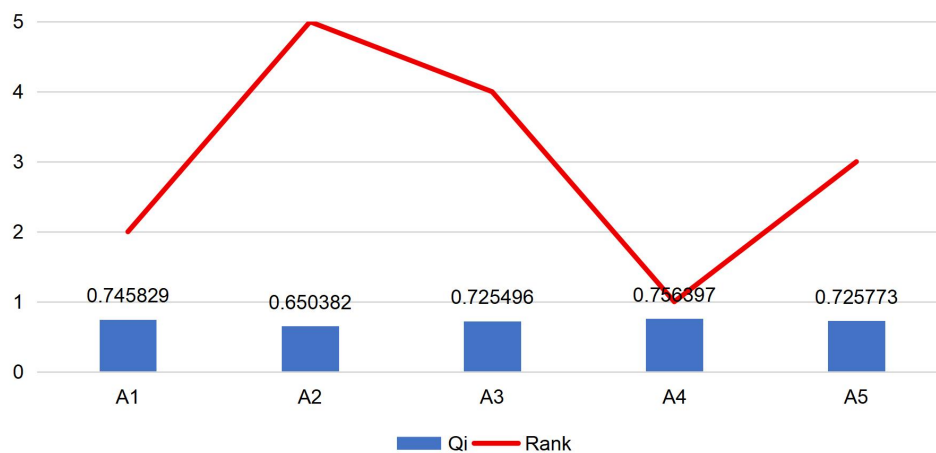


Figure 5. The values of total relative importance and ranking of the alternatives

Based on the results obtained by the WASPAS method, alternative A4 stands out as optimal with the highest value of total relative importance, which indicates its dominant importance in the evaluation and represents the optimal location for a military airport. They are followed by A1 and A5, which occupy stable middle positions and show competitiveness in relation to the leading alternative. Alternative A3 is very close to A5, which indicates almost equivalent performance. The lowest ranked is A2 with the lowest degree of relative importance within the considered

set. The total difference between the optimal and the worst alternative is 0.106015, which represents moderate discrimination and confirms the stability of the rank. The ranks are fully consistent with the Q_i values, without conflict, which further confirms the reliability of the method. For additional validation, a sensitivity analysis of the parameter λ within the WASPAS method, as well as a comparison with other MCDM methods will be performed in the comparative analysis.

4 Validation of Research Results

In order to further validate the proposed methodology, a sensitivity analysis was performed in the following text. The subject analysis was performed on changes in the parameter λ (Table 16).

Table 16. Results of sensitivity analysis

		A1	A2	A3	A4	A5
$\lambda = 0.1$	Q_i	0.763492	0.663527	0.741155	0.777065	0.739192
	Rank	2	5	3	1	4
$\lambda = 0.2$	Q_i	0.759076	0.660241	0.737240	0.771898	0.735837
	Rank	2	5	3	1	4
$\lambda = 0.3$	Q_i	0.754660	0.656954	0.733325	0.766731	0.732483
	Rank	2	5	3	1	4
$\lambda = 0.4$	Q_i	0.750245	0.653668	0.729410	0.761564	0.729128
	Rank	2	5	4	1	3
$\lambda = 0.5$	Q_i	0.745829	0.650382	0.725496	0.756397	0.725773
	Rank	2	5	4	1	3
$\lambda = 0.6$	Q_i	0.741413	0.647096	0.721581	0.751231	0.722419
	Rank	2	5	4	1	3
$\lambda = 0.7$	Q_i	0.736997	0.643810	0.717666	0.746064	0.719064
	Rank	2	5	4	1	3
$\lambda = 0.8$	Q_i	0.732581	0.640524	0.713751	0.740897	0.715709
	Rank	2	5	4	1	3
$\lambda = 0.9$	Q_i	0.728166	0.637238	0.709836	0.735730	0.712354
	Rank	2	5	4	1	3
$\lambda = 1$	Q_i	0.723750	0.633951	0.705922	0.730563	0.709000
	Rank	2	5	4	1	3

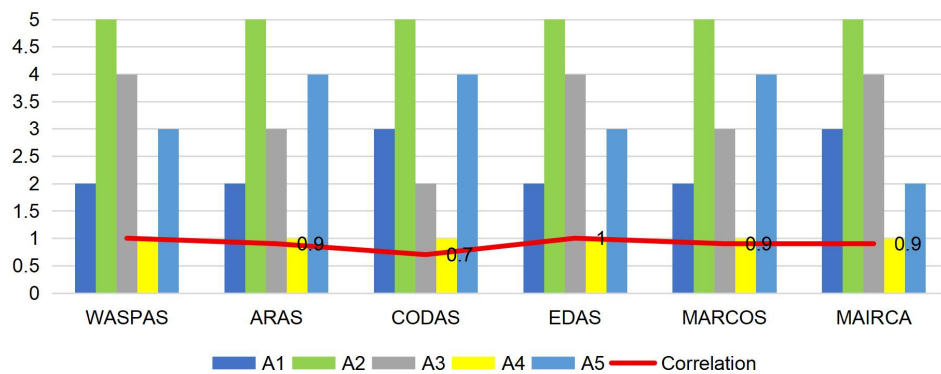


Figure 6. Results of comparative analysis

Note: Weighted Aggregated Sum Product Assessment (WASPAS); Additive Ratio Assessment (ARAS); Combinative Distance-Based Assessment (CODAS); Evaluation Based on Distance from Average Solution (EDAS); Measurement Alternatives and Ranking according to Compromise Solution (MARCOS); Multi-Attributive Ideal-Real Comparative Analysis (MAIRCA)

The sensitivity analysis indicates that the ranking of the alternatives is extremely stable when the parameter λ changes. Alternative A4 is in the first place in all cases, while A2 is always the last, and A1 consistently takes the second place, which indicates a high degree of robustness of the decision. Changes occur only between alternatives A3 and A5: for low values of λ (from 0.1 to 0.3), A3 is ranked better than A5, while at higher values of λ (from 0.4 to 1), their positions are exchanged, so A5 becomes somewhat more favorable. The Q_i values of all alternatives gradually decrease with the increase of λ , but the relationships among the alternatives remain almost the same, which further confirms the stability of the ranking. The only critical point is the transition between $\lambda = 0.3$ and $\lambda = 0.4$,

where the ratio between A3 and A5 changes, but this does not affect the final decision, because the optimal and the worst alternative do not change in any scenario. The results show that the choice of the value of λ has no significant impact on the final ranking in this decision problem, which makes the WASPAS method a reliable tool for selecting the optimal location for a military airport.

In addition to the sensitivity analysis, a comparative analysis was also performed. The comparison was made with five MCDM methods: Additive Ratio Assessment (ARAS), Combinative Distance-Based Assessment (CODAS), Evaluation Based on Distance from Average Solution (EDAS), Measurement Alternatives and Ranking according to Compromise Solution (MARCOS), and Multi-Attributive Ideal-Real Comparative Analysis (MAIRCA). To determine the correlation between ranks, Pearson's correlation coefficient was used [50]. The results of the subject analysis are presented in Figure 6.

The results of the comparative analysis shows that the WASPAS method has a high correlation with ARAS, MARCOS and MAIRCA, while EDAS achieves the maximum correlation, which indicates almost identical ranks and values between those methods. In contrast, CODAS shows a slightly lower correlation, which suggests deviations in ranking and a slightly different approach to evaluation. As with the sensitivity analysis, alternative A4 is optimal in all methods. This analysis confirms that the WASPAS method is consistent and reliable, with results that are in close agreement with most other MCDM methods with which it is compared.

5 Conclusions and Future Research

The choice of a location for a military airport is a decision of strategic importance, as it affects the security of the country, the efficiency of military operations, and the successful long-term planning of the defense infrastructure. The complexity of this problem arises from the need to simultaneously consider numerous criteria, which makes it suitable for the application of MCDM methods. The results of the conducted research confirm that hybrid MCDM models allow a systematic approach to such problems, especially when the criteria differ in nature, importance, and degree of uncertainty.

Within the framework of the proposed methodology, the criteria weights were determined using the DIBR II method and its fuzzy, grey and rough variants, which enabled more realistic modeling of subjective assessments and missing information. Aggregation of expert opinions was implemented using the EWAA operator, which allows consideration of different levels of expertise and reduces the influence of extreme values. The WASPAS method was used to rank the alternatives, which showed a high degree of stability. According to the results obtained, alternative A4 stood out as the most favorable, while A1 and A5 occupied intermediate positions, and A2 proved to be the least suitable. The difference between the optimal and the worst alternatives indicates a moderate but clear difference in their performances.

The stability of the model was additionally checked by analyzing the sensitivity of the parameter λ . Changes in the value of this parameter did not affect the identification of the optimal and worst alternatives, which indicates a high level of robustness of the decision. The only changes were observed between alternatives A3 and A5, but they did not affect the final choice. These results confirm that the WASPAS method works reliably even in conditions of changing aggregation structure.

Comparative analysis with five other MCDM methods further confirmed the consistency of the model. A particularly high correlation with the EDAS, MARCOS and MAIRCA methods indicates that the ranks obtained by the WASPAS method match the results of other established approaches. In all methods, alternative A4 remains optimal, which additionally confirms the validity and stability of the proposed model.

Further research development can be directed in several directions. Integration of GIS tools with the MCDM model would enable spatial analysis and visualization of the terrain, which would further improve the accuracy of location selection. The application of more advanced fuzzy environments, such as spherical or q-rung orthopair fuzzy sets, could further improve uncertainty modeling. Also, the integration of machine learning methods could contribute to the automation of weighting or scenario prediction, and the inclusion of a larger number of experts from different institutions would increase the objectivity and applicability of the decision made.

Although the proposed hybrid MCDM framework provides a reliable basis for military airport location selection, certain limitations must be considered. The criteria used are representative, but not exhaustive, and specific operational or geopolitical circumstances may require additional parameters. The number of alternatives analyzed is limited, which may affect the breadth of insight. Although sensitivity and comparative analyses were conducted, the model was not tested in simulation scenarios that would demonstrate its behavior under extreme conditions, and the results are context-specific and cannot be directly generalized without adaptation, although the methodological framework remains broadly applicable.

Author Contributions

D.T. and D.B. jointly designed the research, developed the methodology, collected and analyzed the data, and wrote and revised the manuscript. The authors read and approved the final version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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